INVERTIS UNIVERSITY, BAREILLY

FACULTY OF ENGINEERING & TECHNOLOGY

MID-SEMESTER EXAMINATION 2023-24

Academic Session: 2023-2024

Programme: B.Tech. (Computer Science & Engineering) | **Semester: Semester III**

Course Code: BCS-301 | **Course Title: Data Structures**

Time Allowed: 1.5 Hours | Maximum Marks: 25

Note: Attempt all sections as per instructions. Assume missing data suitably, if any.

SECTION A (Short Questions - 2 Marks each)

Q1(a). Calculate address of element A[3][4] in row-major order with base address 1000.

Q1(b). Write the prefix form of the infix expression: (A + B) * (C / D) - E.

Q1(c). State two applications of Queue in operating systems.

SECTION B (Detailed Questions - 5 Marks each)

Q2. Write an algorithm to reverse a singly linked list without using auxiliary storage.

Q3. Implement two stacks using a single one-dimensional array without overflow.

SECTION C (Comprehensive - 9 Marks)

Q4. Explain Circular Queue implementation using arrays. Write functions for Enqueue and Dequeue.

========== END OF QUESTION PAPER ==========